Men's Classified Ads and Risky Health Behaviors

By HASAN SHAHID*

Risky health behaviors are private and rarely observed. As a result, researchers studying behaviors such as risky sex have largely relied on self reported survey data or outcomes data, such as rates of sexually transmitted infection. Both approaches provide limited insight. Survey responses are often subject to systematic misreporting, while outcomes data obscure the specific behaviors underlying observed changes.[1]

The early years of the HIV/AIDS epidemic represent a period in which individuals may have perceived a substantial increase in the costs associated with risky sexual behavior, even as reliable information about transmission and prevention remained limited. This paper contributes new descriptive evidence by constructing a novel dataset of over 170,000 mens classified advertisements published between 1975 and 1992 in *the Advocate*, the leading national LGBT magazine in the United States during this period. In this setting, classified advertisements primarily consist of men seeking other men for romantic and sexual relationships. A few years into the HIV/AIDS epidemic, I observe significant changes in mens classified advertisements. Specifically, the total number of ads declines while the proportion of ads that mention safe sex rises. Together, these patterns document meaningful changes in participation and language use within a market that offers a rare window into sexual behavior during the early years of the epidemic.

This paper relates to a broader literature examining behavioral responses to changes in the perceived costs of risky sexual activity. Some studies in this literature exploit variation in sexually transmitted infection risk (Oster, 2005; Auld, 2006; Shahid, 2024; Spencer, 2024), while others focus on changes in information that shape perceptions of risk (Kerwin, 2020; Delavande and Kohler, 2012). My contribution differs from this work by using high frequency historical data to document specific behavioral responses, rather than relying primarily on outcomes data.

I. Data and Setting

The early years of the HIV/AIDS epidemic represent a unique setting to study behavioral responses to an increase in the perceived cost of risky sexual behavior. In the summer of 1981, news reports began to describe a previously unknown illness affecting gay men in New York and California. Although the underlying cause of the illness was not yet understood, early coverage frequently associated it with frequent sexual encounters among gay men (Altman, 1981). Scientific evidence identifying the illness as being caused by a virus, later named Human Immunodeficiency Virus, did not emerge until 1983. Nonetheless, community level responses preceded this discovery. Well before the virus was identified, gay community organizations promoted behavioral changes that emphasized reducing sexual activity and adopting safer sex practices.[2]

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[1] For example, self reported sexual behaviors may be misreported in non random ways, and changes in STI rates can reflect shifts in testing, treatment, or reporting rather than behavior alone (Corno and De Paula, 2019).

[2] In June 1982, the San Francisco based organization *The Sisters of Perpetual Indulgence* distributed a flier titled "Play Fair" that encouraged safe sex practices (Gay in the 80s, 2013). In July 1982, the New York based organization *Gay Men's Health Crisis* released its first newsletter, which included recommendations to limit the number of sexual partners to reduce the spread of AIDS (Terra, 2023).

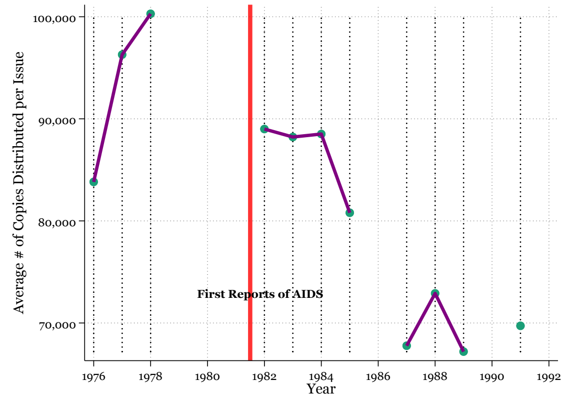

FIGURE 1. COPIES DISTRIBUTED PER ISSUE

Source: Advicate Classified Ads: This figure presents distribution statistics from located "Statement of Ownership, Management & Circulation" from issues of *the Advocate*. The vertical axis represents the average number of copies distributed per issue over the past 12 months.

The Advocate remained the most popular national LGBT publication accounting for approximately 10% of the combined circulation of all lesbian and gay publications (Streitmatter, 1995). Although comprehensive circulation records are unavailable, some information can be recovered from "Statements of Ownership, Management and Circulation" published in issues of the magazine. These statements report the average number of copies distributed per issue over the previous twelve months. Figure 1 presents distribution statistics in the located statements.

Unlike many other LGBT publication, *the Advocate* was unique with significant circulation in many U.S. cities. *The Advocate* has included a classified section since 1967.[3] Advertisers submit text by mail and pay a fee based on length.[4] While classifieds are used for multiple purposes, the overwhelming majority consist of men seeking other men for romantic or sexual relationships.

The primary dataset is constructed by digitizing issues of *The Advocate* purchased by the Vanderbilt University Library through ProQuest. Classified advertisements appear sequentially in column format across multiple pages of the publication. I exploit the vertical column structure of the advertisements identified using variation in the proportion of dark pixels across each page together with Optical Character Recognition to construct a dataset in which each observation corresponds to the text of an individual advertisement. Since the classified columns also contain general business advertisements, as well as postings related to employment, housing, or tenants, I apply a machine-learning classifier to remove all entries that are not placed by an individual seeking a social, romantic, or sexual connection with another individual. This process yields a dataset of 176,906 ads across my years of observation.

I analyze the text of each advertisement to extract information about advertiser characteristics and preferences. Disclosure is limited by the short length of ads, with a median of 25 words, though many advertisers report basic demographics such as age and race. One challenge in identifying demographic information is that when a particular race is mentioned, it is often unclear whether it refers to the advertiser or to a preferred partner. To address this ambiguity, I identify verbs within the ads that allow me to distinguish between self-descriptive text and descriptions of preferred partners.[5]

Beyond demographic information, I categorize ads along economically meaningful dimensions.

[3]The earliest issue of *The Advocate* available through Proquest was published on September 1, 1967 and contains a classified section.

[4]Fees increase with the number of characters included in an advertisement.

[5]For example, text following verbs such as "seeks" or "looking for" is classified as describing a preferred partner, while text preceding those verbs, as well as text in other sentences, is interpreted as describing the advertiser. Similarly, when verbs such as "wanted" or "desired" appear, text preceding those verbs is classified as describing a preferred partner, with all remaining text interpreted as describing the advertiser.

TABLE 1—SUMMARY STATISTICS OF CLASSIFIED ADS

	Mean	Std. Dev.
Self Race - White	0.877	0.328
Self Race - Black	0.063	0.243
Self Race - Asian	0.032	0.176
Self Race - Not Reported	0.375	0.484
Self Age	31.495	10.699
Self Age Not Reported	0.505	0.500
Long Term Relationship Oriented	0.307	0.461
Explicitly Sexual	0.728	0.445
Safety	0.069	0.254
State – California	0.458	0.498
State – New York	0.246	0.431
State – Illinois	0.036	0.187
State – Not Identified	0.288	0.453
Observations	176,901	

Advocate Classified Ads: This table presents summary statistics from the all ads posted in the Advocate from 1975-1992.

Using natural language processing, I classify whether an ad expresses a preference for a long-term relationship or is explicitly sexual. I randomly select 300 ads and manually label them along these dimensions, which I use as training data for a supervised machine learning model that is then applied to the full sample.[6] I also identify ads containing language related to safety (these include the terms, "safe," "condom," or "protect"), which I classify as safety ads. This category captures behavioral responses to AIDS, including the adoption of safer sex practices.[7]

The ads also include several geographical identifiers. For example, most ads include phone numbers and the first three digits of that phone number represent an area code. The ads are also arranged by state, such that if the ads immediately above and below a given ad are from the same state, the intervening ad is inferred to be from that state as well.

I provide a summary of these statistics in Table 1.[8] The majority of ads are from young white men. Approximately 73% of ads are explicitly sexual and 31% of the ads express a preference for long-term relationships. 7% of ads mention some safety term. I also present statistics on the top three states represented in the classified section. I am able to identify the advertiser's state for approximately 70% of the ads. California, New York, and Illinois have the greatest representation in the classifieds.

II. Time Series Evidence

In Figure 2, I present the total number of classified ads in my dataset in each issue overtime. The lighter line represents the exact number of ads in each issue while the darker line represents a smoothed version obtained from a locally weighted regression. In general, there is a large variation in the total number of ads posted per issue where most issues have anywhere between 350 and 750 ads. I also find that the number of ads posted exhibits strong serial correlation, with counts in one issue closely related to those in the prior issue. Overall, the average number of ads increases in the years leading up to the first reports of AIDS in mid-1981, with a further rise in the early years of the epidemic. The number of ads peaks in late 1983, followed by a sharp decline, after which most issues

[6]I use Term Frequency–Inverse Document Frequency (TF-IDF) to vectorize the ads. TF-IDF downweights words that are common across documents and performs better than word-embedding approaches in this context, as classified ads frequently rely on slang, coded language, and abbreviations that are poorly captured by pre-trained embeddings.

[7]I do not use supervised machine learning to classify safety ads given the heterogeneity of language used in these contexts.

[8]All demographic characteristics (race and age) refer to the advertiser rather than the preferred partner.

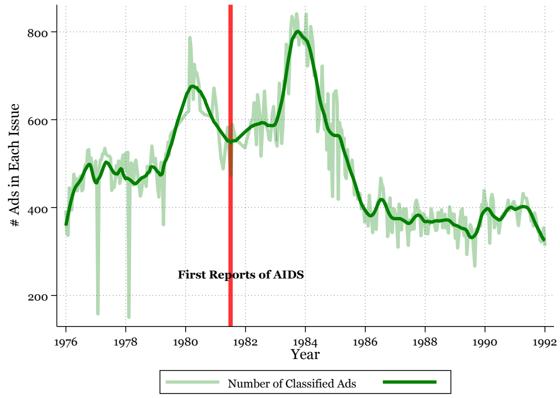

FIGURE 2. TOTAL ADS OVER TIME

Source: Advocate Classified Ads: This figure represents the total number of ads posted in each issue of *the Advocate* personals section over time. The light line represents the exact number of ads in an issue while the darker line represents a smoothed version obtained from a locally weighted regression.

contain roughly 400 ads.[9]

Although the decline in the total number of classified ads observed a few years into the epidemic is notable, simple trends in overall ad volume may reflect multiple contributing factors. Specifically, the total number of ads we observe in the classifieds are the result of both demand and supply side factors. While demand-side factors include behavioral responses to the epidemic, supply-side factors reflect policy changes implemented by the magazine. For example, individuals pay a small fee for their ads to appear in a future issue of the *the Advocate*. Overtime, this fee has increased.[10] Additionally, while the overall format of the ads remains consistent for most of the period, there are several changes in the subsections which appear in the classifieds.[11] An additional limitation relates to the construction of the advertisement level dataset. In some issues, dense column formatting makes it difficult to determine where one advertisement ends and another begins, which may result in double counting a single advertisement or counting multiple advertisements as one. The severity of this issue may vary over time. Taken together, these considerations suggest that while changes in the total number of classified advertisements are informative, they should be interpreted with caution when viewed as reflecting behavioral responses to HIV/AIDS related risk.

In Figure 3, I present time series patterns in the proportion of advertisements that include safety related language. References to safety are nearly absent prior to the first reports of HIV/AIDS in the summer of 1981. In the years immediately following these reports, the prevalence of safety language increases modestly, followed by a more pronounced rise in the mid-80s. Safety related advertisements peak around 1988 and then decline thereafter. This suggests that the early years of the HIV/AIDS epidemic were associated with a large increase in the use of safety related language in classified advertisements.

III. Conclusion

Taken together, the time series evidence indicates meaningful behavioral changes during the early years of the HIV AIDS epidemic, including a substantial increase in the use of safety related language in classified advertisements. More broadly, these findings highlight that in settings where conven-

[9]This rise prior to the first reports of AIDS and the subsequent decline in the mid-1980s closely mirror trends in the total number of issues distributed per year depicted in Figure 1.

[10]For example, the price of an 81 character advertisement increased from approximately five dollars in the late 70s to about sixteen dollars in the mid 80s, representing a substantial rise even after adjusting for inflation.

[11]For example, from May 1983 to August 1984, there is a “Women’s Ads” subsection which does not appear in future issues.

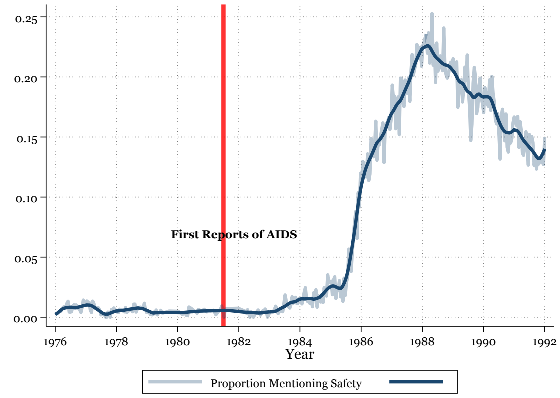

FIGURE 3. PROPORTION OF SAFETY ADS OVER TIME.

Source: Advocate Classified Ads: This figures represent the proportion of ads which include the terms 'safe', 'condom', or 'protect' in each issue of *the Advocate* classified section over time. The light line represents the exact proportion of ads in an issue while the darker line represents a smoothed version obtained from a locally weighted regression.

tional data sources are limited, particularly when studying minority populations that are not readily observed in standard datasets, unconventional sources such as historical newspapers and magazines can provide valuable insight into patterns of behavior.

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